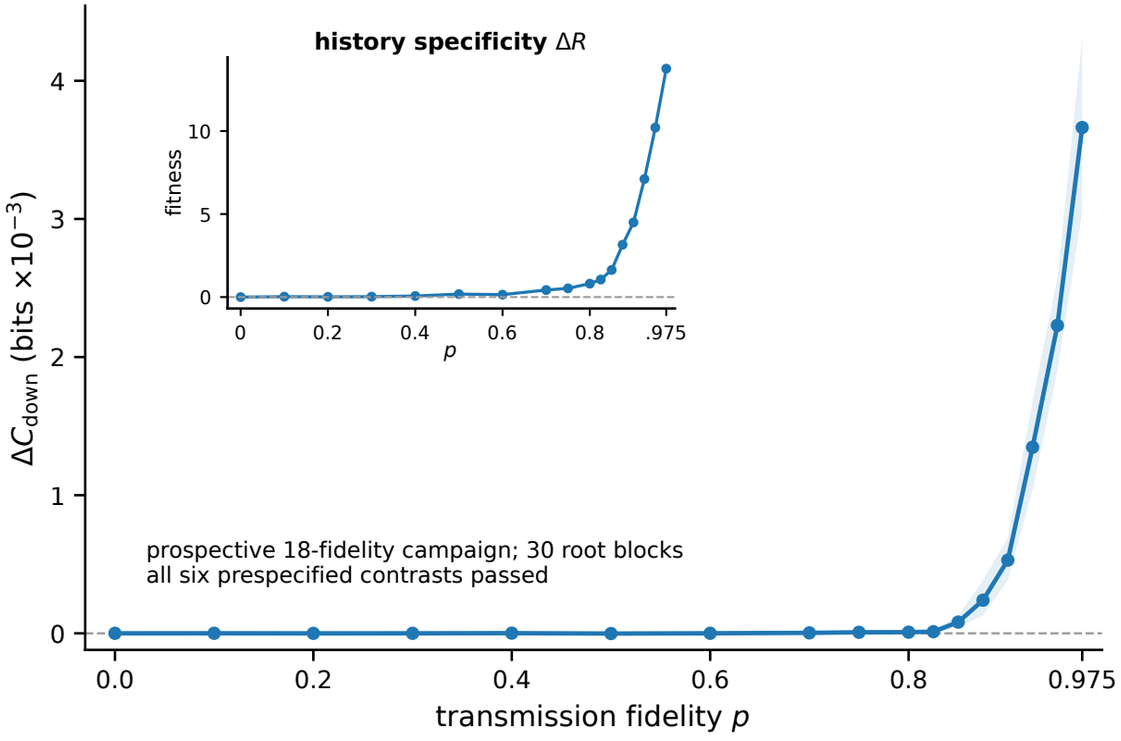
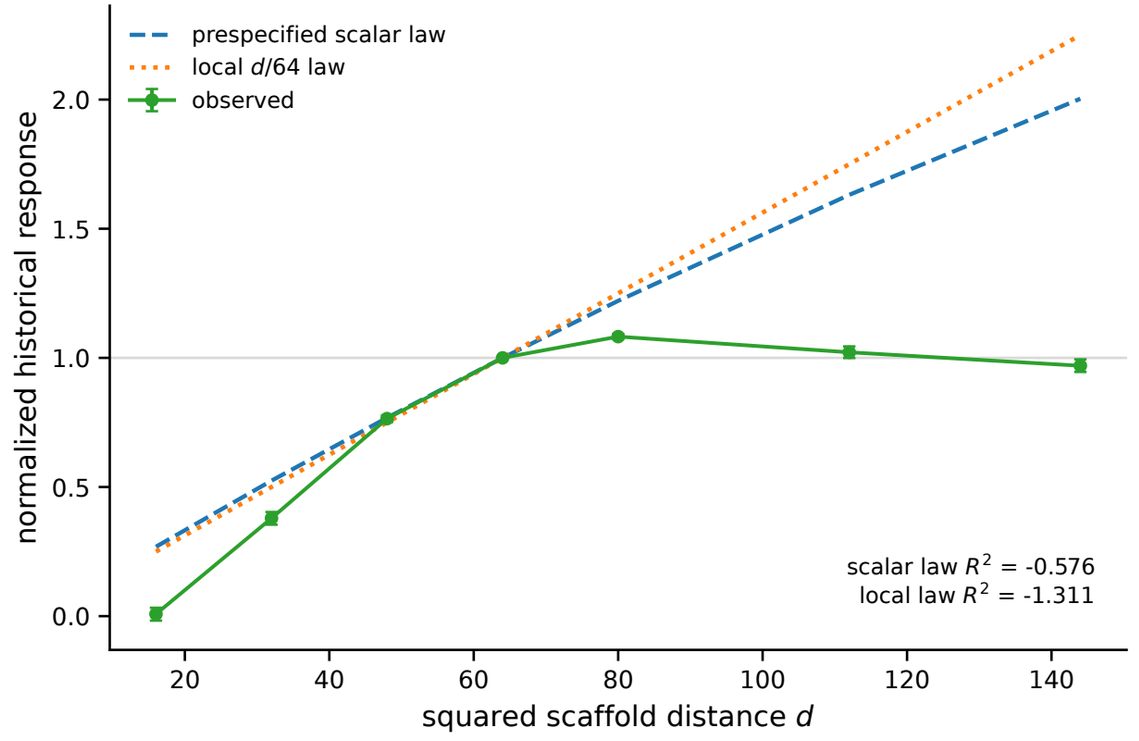
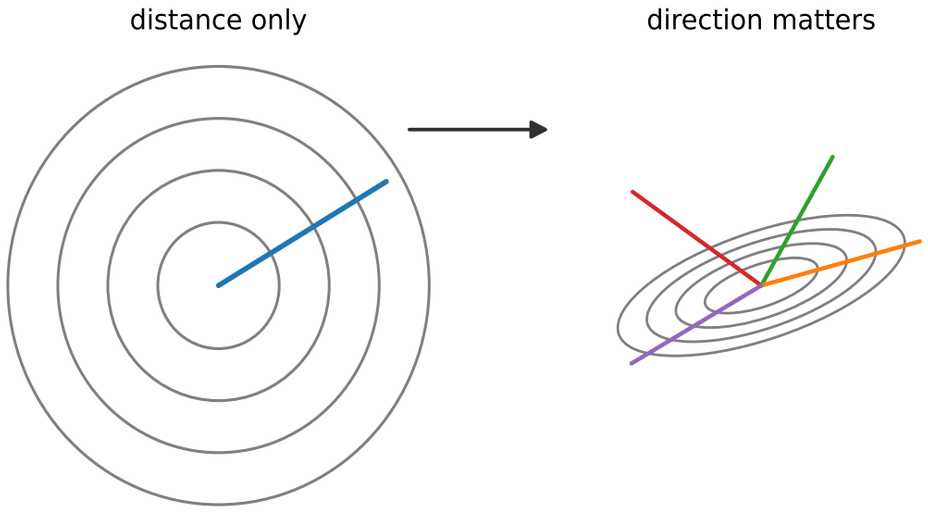
(a) **Reliable heredity amplifies inherited causal organization**



(b) **Scalar environmental distance fails prospectively**



(c) **Directional response replaces a distance-only law**



$\mathcal{R} = f(\|\delta\|)$ $\mathcal{R}(B_H + \delta) \approx R_0 + g_T^T\delta + \frac{1}{2}\delta^T H_T \delta$

Prospective failure of the scalar-distance law motivated the revised local geometry, which permits anisotropic response.

(d) **Equal-distance directional profile is predicted**

